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Rethinking urban forests as essential infrastructure for resilience, equity, and biodiversity in the current climate emergency

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Abstract

Urban forests are not merely green amenities; they support critical ecosystem functioning and services vital for healthy, resilient cities. Recognising urban forests as core infrastructure is essential to reversing the loss of mature trees, preserving biodiversity,

and maintaining liveability amid increasing climate and environmental pressures. Although the benefits of urban forests for climate resilience, biodiversity, and public health are broadly acknowledged, policies to protect and enhance these vital ecosystems are often limited, underfunded, and inadequately enforced. As mature canopy loss today takes decades to be replaced (if ever), immediate and sustained investment is crucial to safeguard urban forests. This urgency reveals four interconnected gaps in current urban forest management and stewardship. First, urban forests require recognition, investment, and maintenance as essential infrastructure contributing to urban resilience, including biodiversity support, and to maximise the delivery of key ecosystem services such as cooling and carbon sequestration. Second, equitable access to greenspaces across all communities must be ensured to redress long-standing social and environmental injustices. Third, integrating urban forests into broader climate and biodiversity governance frameworks is critical to mainstreaming their management and protection. Lastly, resilience must be strengthened through evidence-based management practices responsive to evolving environmental changes and social contexts. These priorities must be complemented with strong legal protections, rigorous enforcement of legislation against illegal tree removal, and robust community engagement supported by integrated urban planning and improved monitoring. Without these, the ecological, social, and economic benefits provided by urban forests will remain threatened. By reframing urban forests as essential living infrastructure embedded in legal, financial, and planning frameworks, cities can become cooler, healthier, more biodiverse, and socially just. This framework offers timely guidance for policymakers to prioritise urban forests within climate resilience and sustainability strategies, securing benefits for current and future generations.

Introduction

By 2050, nearly 70% of humanity will reside in cities [1], intensifying demand for greenspaces and the vital ecosystem services they provide [2,3]. Cities stand at the frontline of the climate crisis, concentrating both risks and opportunities for transformative action [4]. Rapid urban expansion drives greenhouse gas emissions, alters local climates, and accelerates biodiversity loss but also provides new opportunities for systemic change [4–6]. Recognising urban forests as key components of low-carbon, climate-resilient development highlights the urgency of embedding them in global and local policy frameworks [7].

Urban forests encompass all trees and woody vegetation in urban landscapes, from city centres to peri-urban areas. They range from forests and woodlands, which are characterised by trees and understory vegetation, to groups of individual trees found along streets and in parks, gardens, private and public lots [8]. Trees on public and private properties and urban woodlands (planted, spontaneous, and remnant) provide crucial ecosystem services (i.e., benefits that humans receive), which sustain and enhance cities' climate infrastructure. Notably, many classification systems differentiate 'forest' as areas with relatively closed canopies from 'woodland', which refers to more open, scattered tree formations. Much of the tree-rich urban landscape falls within this woodland-like continuum. While we use 'urban forest' as the established term, recognising this spectrum can enhance ecological framing and planning for resilient urban greenspaces.

Urban forests are not just green amenities. As a dominant feature of green infrastructure [9,10], they deliver multiple ecosystem services, including cooling urban temperatures, reducing pollution and noise, enhancing soil infiltration, slowing stormwater runoff, providing habitat, protecting from UV radiation, supporting social recreation, and providing culturally important spaces and resources, all leading to a better quality of life [10–20]. Urban forests also contribute to climate adaptation and mitigation by buffering weather extremes, sequestering carbon, and lowering energy demand for cooling [21–23]. They promote healthier lifestyles, reduce stress-related illnesses, enhance wellbeing, and can improve local food security and health through urban foraging [24–28]. Cultural ecosystem services, such as recreation, aesthetics, and spiritual enrichment, alongside traditional worldviews of nature, are gaining even more importance amid the climate emergency [29,30]. When managed effectively, urban forests increase in value over time and their benefits far exceed costs to maintain the tree resource [3,12,13].

Trees in urban settings, however, face increasing pressures such as drought and heat stress, soil compaction, restricted rooting volume, invasive pests and diseases, pollution, wounding, and sometimes vandalism, all of which can lead to reduced growth or even mortality [31–34]. These pressures, driven by climate change and urban expansion, as well as weak legal protection and inadequate funding, threaten ecosystem services and resilience (Fig 1) [21,35–42]. Furthermore, these threats can be even higher in areas with low species diversity, which is common in urban landscaping, and make urban forests more vulnerable to pests, pathogens, and environmental stressors [42–44]. Additionally, urban forests dominated by trees concentrated in the same age or size cohort compound the risks of canopy loss, structural decline, and synchronous mortality events [45]. Promoting both taxonomic, genetic, functional, and structural diversity is therefore essential to maintain resilience and ensure the long-term stability of urban forest ecosystems.

While the benefits of urban forests are widely recognised, they are not without disservices. For example, certain tree species can negatively affect urban inhabitants by releasing allergens that trigger respiratory issues [46–48], by emitting biogenic volatile organic compounds (BVOCs), or via inappropriate planting locations, which can contribute to localised urban air pollution [49,50]. In some instances, expansive or shallow root systems of some tree species can interfere with pavements, water pipes or other adjacent infrastructure, resulting in structural damage over time [51,52]. Addressing these risks requires inclusive, data-driven planning that aligns ecological goals with local community needs and preferences. Urban forest planning must incorporate thoughtful species selection in conjunction with local communities and

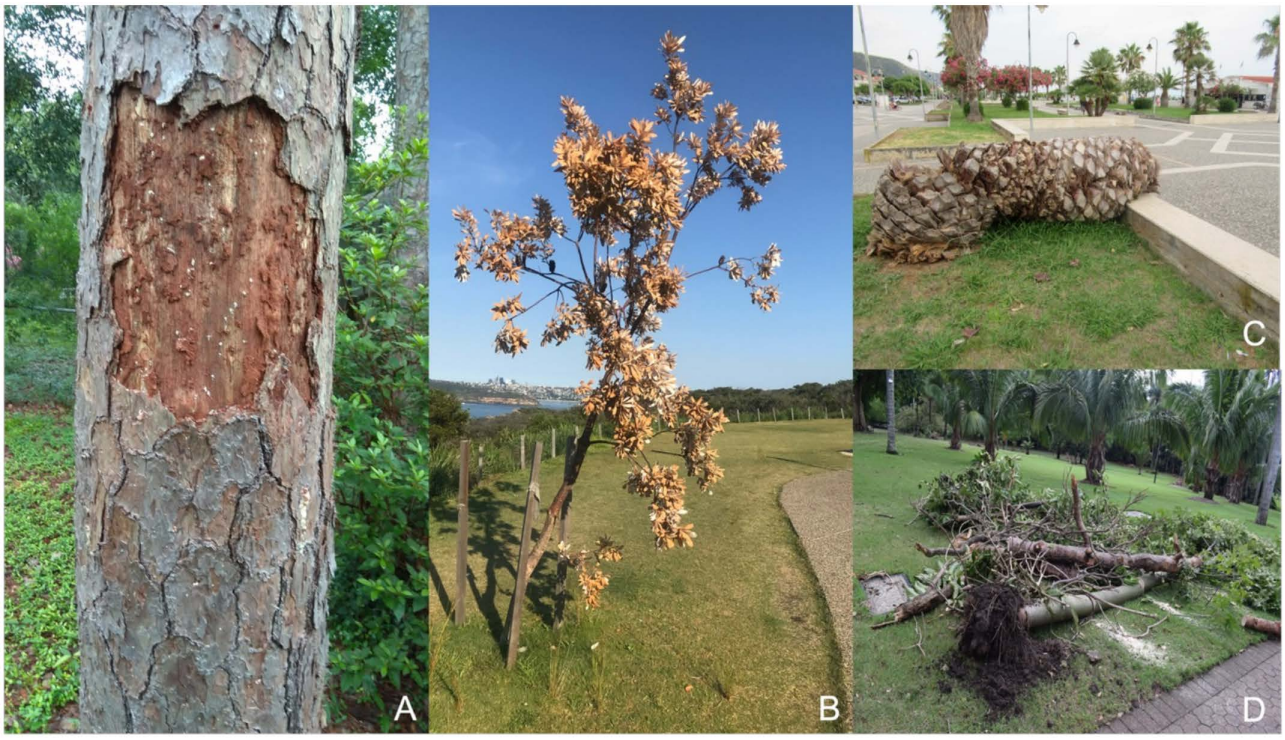


Fig 1. Examples of urban tree vulnerabilities to extreme weather events or pest/disease outbreaks events across the globe. (A) Loblolly pine tree (*Pinus taeda*) killed by a bark beetle in Tallahassee, FL, USA; (B) *Banksia* spp. dieback after an extreme heatwave and drought event in Sydney, Australia; (C) palm tree collapsed due to infestation by *Rhynchophorus ferrugineus*, in Ascea Marina, Salerno, Italy; and (D) fallen tree caused by Severe Tropical Cyclone Alfred in Brisbane, Australia. These examples illustrate that, without proactive management and policy support, urban trees worldwide remain at increased risk from biotic and abiotic stressors. Applying policy priorities, such as strengthening pest and disease monitoring, selecting climate-resilient and diverse species, enforcing tree protection regulations, and investing in adaptive maintenance, can reduce future failures and help safeguard urban forests as essential infrastructure. Photos provided by EL (A), MER (B), and AR (C and D).

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equitable spatial design, supported by appropriate tree management, to minimise disservices. Balancing these considerations with the goal of preserving and restoring urban ecosystems is essential for effective planning and sustainable outcomes.

The complex trade-offs between the benefits and costs of urban forests, as well as the effects of policy interventions on ecosystem services, are not yet fully understood [53,54], although the negative consequences of canopy loss are well documented [55–61]. Notably, canopy losses are often most acute in densely populated neighbourhoods where benefits are most needed [62]. Minimising these declines requires local and site-specific approaches that ensure appropriate species selection based on site conditions, climate and local preferences and the implementation of policies that conserve both remnant and newly planted urban forests. Moreover, inequitable access to urban forests exacerbates other social and environmental injustices and increases health disparities [63–69]. In several countries, low-income and marginalised communities can often be characterised by low canopy cover, limited proximity to green spaces, and greater exposure to the adverse impacts of urban heat and air pollution [63–66,70]. These patterns contribute to disproportionate risks of heat stress, poor air quality, and fewer opportunities for recreation and wellness benefits and provisioning services [63]. As a result, prioritising urban forestry policies that enhance equitable access to well-managed green spaces and target tree-planting interventions in underserved areas is essential for promoting social equity and maximising the public health benefits of urban ecosystems [71].

In addition to strong policy support, urban forest management requires continuous monitoring and community engagement. It should follow inclusive local and site-specific approaches that balance biodiversity, ecosystem function, and social needs and preferences, supported by quantitative and achievable canopy and diversity targets [72–74]. Many cities are now developing quantitative, neighbourhood and local targets for urban forests that go beyond broad city-wide canopy goals [72,75–77]. This shift can ensure that neighbourhoods with low canopy cover, where residents lack access to established green areas and their benefits, are not overlooked within larger spatial scale targets [63]. Importantly, protecting mature trees is essential, since newly planted trees cannot replace lost canopy in the short term and can take decades to provide similar ecosystem services. This requires continuous investment, robust legal frameworks, enforcement mechanisms, and site-specific management approaches that recognise the heterogeneity of urban forests [74].

This essay presents a framework advocating for the positioning of urban forests as essential living infrastructure supporting climate resilience, biodiversity, and wellbeing (Fig 2). We identify four policy priorities—investment in infrastructure, equitable access, mainstreaming governance, and evidence-based management—and discuss complementary actions for effective stewardship, legal protections, and economic incentives. Ultimately, the goal is to guide policymakers in embedding urban forests within broader sustainability and resilience strategies to benefit current and future generations.

Policy priorities for immediate action

1. Investing in urban forests as essential living infrastructure for health and resilience

Recognising urban forests as essential living infrastructure requires their integration into legal, financial, and planning frameworks, including municipal ecosystem accounting and city council budgeting processes [78,79]. Many cities (although yet not enough) are already moving beyond tree planting targets to adopt site- and function-specific strategies, aiming to retain and enhance existing green infrastructure rather than relying on a “clear and build” development approach [e.g., 78–80].

Equally important is safeguarding and enhancing soils, understory vegetation, and habitat, often overlooked elements critical for long-term tree health and ecosystem functioning [81–84]. Healthy soils support root growth, nutrient cycling, water infiltration, and pollutant filtration [85,86], forming the foundation for resilient urban forests. Urban forestry initiatives should thus adopt holistic approaches integrating soil management, habitat diversity, and species-rich planting to deliver equitable ecosystem services across communities.

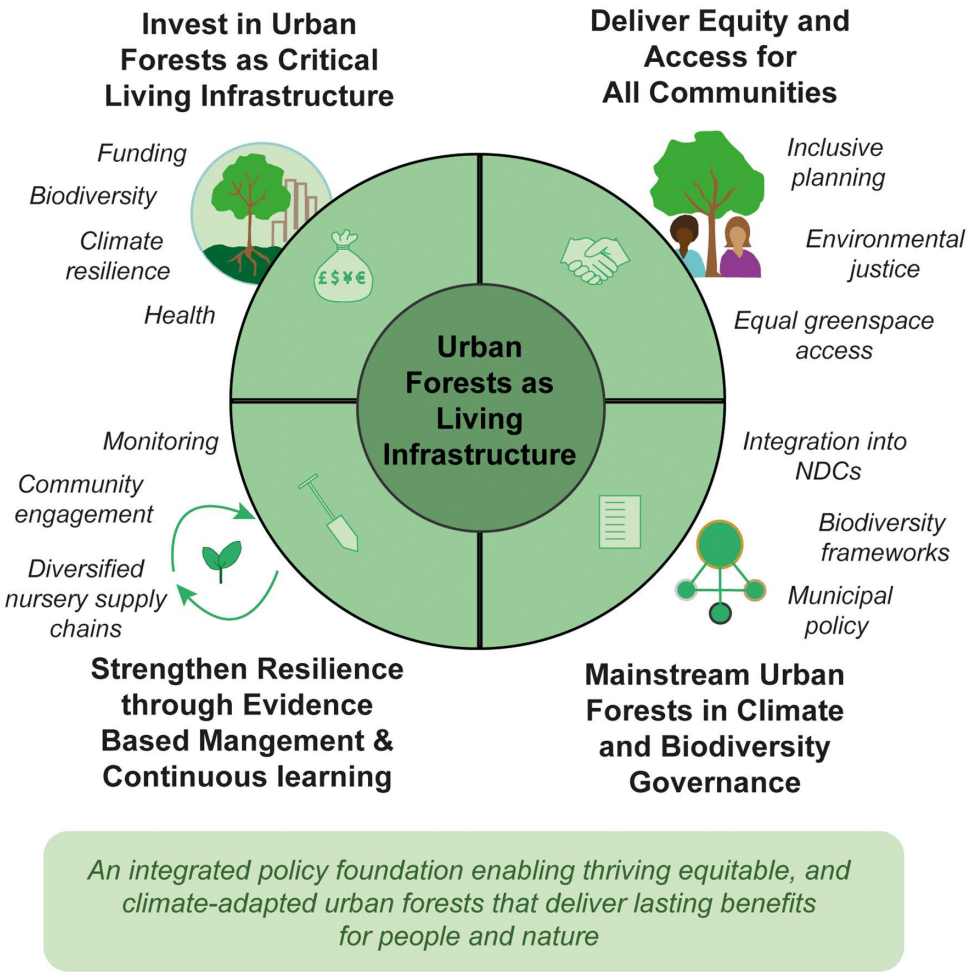


Fig 2. Schematic diagram illustrating the four interconnected and complementary policy priorities essential for safeguarding and enhancing urban forests. These include: (1) investing in urban forests as essential and critical living infrastructure for climate resilience and biodiversity; (2) delivering equity and access to ensure all communities benefit from urban forests and greenspaces; (3) mainstreaming urban forests within global and local climate and biodiversity governance frameworks to secure sustained commitments and financing; and (4) strengthening resilience through adaptive management supported by monitoring, community engagement, and diversified nursery supply chains. The framework emphasises the need for integrated legal, financial, and planning mechanisms that recognise urban forests as living infrastructure and critical public assets requiring long-term stewardship. This policy foundation supports thriving, equitable, and climate-adapted urban forests that deliver multiple benefits to current and future urban residents. Image generated by ChatGPT (OpenAI, 2025).

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Equating urban forest stewardship with other core urban infrastructure, such as drainage and sewer systems, roads and bridges, and electricity and water supply networks, is essential to reversing losses of valuable mature trees, creating viable planting spaces, preserving biodiversity, and ensuring future cities are liveable and healthy. Consequently, municipal budgets should incorporate urban forest management plans, allocating resources for staffing and long-term maintenance, as they do for built infrastructure. This mindset shift matters because it can help in removing the silos and prioritisations currently relegating urban forest management to the background, and can create opportunities for dedicated funding streams, enabling more strategic and sustained investment in urban forestry. As a result, such initiatives facilitate improved maintenance, expansion of canopy cover, and urban resilience.

A related and critical challenge that must be acknowledged within such frameworks is the strong profit motive driving some landowners to prioritise land sale and development over urban forest preservation. This economic imperative presents a persistent challenge, as landowners, regardless of parcel size, hold primary responsibility for tree stewardship on their property. Recognising that protecting urban nature constitutes a valuable community service, policy frameworks should combine regulatory measures with meaningful economic incentives. This includes penalising illegal tree removal and damage (e.g., poisoning), while also rewarding preservation and enhancement. While fully compensating landowners for unavoidable development profits may be unrealistic, legislation could mandate minimum levels of urban nature provision on private lands, with enforceable penalties for non-compliance. Complementary measures, such as tax reductions for land supporting permanent tree cover, could further encourage stewardship. Together, these approaches frame tree care as both a legal obligation and a shared social responsibility, balancing economic interests with ecological outcomes. Incentives for private property owners to conserve trees can further augment conservation efforts [87].

Targeted recommendations for enhancing urban forest conservation and planning are summarised in **Box 1**.

2. **Deliver equity and access**

Closing gaps in urban tree provision in underserved neighbourhoods requires setting locally appropriate, site-specific canopy cover, biodiversity and access targets, co-designed with communities to reflect their needs and values [63,88,89]. Public engagement campaigns must actively involve all communities, incorporate cultural perspectives, and deliver outreach in multiple languages [90–93]. Empowering marginalised populations through meaningful inclusion in planning, decision-making, access to resources, and promoting stewardship partnerships is essential to preserve urban forests from climate stressors, pests and pathogens, and inappropriate development [94]. Furthermore, incorporating Indigenous Peoples' and local communities' perspectives ensures equitable stewardship that better respects Indigenous Cultures, Knowledge and Rights, promoting inclusive and just urban forest management [71,95–97]. Targeted planning and resource allocation must affirm greenspace as a fundamental right and diversify the urban forestry workforce to improve representativeness and equity [63,64].

Strategic planning should extend from dense urban centres, often dominated by impervious surfaces and limited or no tree cover [98], to peripheral neighbourhoods. Low-income and marginalised areas often disproportionately experience limited canopy and biodiversity, exacerbating heat and pollution exposure [99–102]. Integrating urban forests within city centres and corridors can enhance climate resilience and social cohesion, reduce disparities, and extend green benefits to all residents [103]. To avoid green gentrification, municipal greening initiatives must coordinate with housing and urban planning departments [98].

Key recommendations to enhance urban forest equity and access are summarised in **Box 2**.

3. **Mainstream urban forests in climate and biodiversity governance**

Effective policies rely on strategic, high-level targets. Thus, urban forests must be integrated across global, national, regional, and local climate mitigation, adaptation, and biodiversity conservation strategies, supported by collective climate finance mechanisms. Linking urban forestry efforts with global governance frameworks, such as the United Nations Framework Convention on Climate Change, the Paris Agreement, the Convention on Biological Diversity, and the UN Sustainable Development Goals [104–106] can help ensure that local actions contribute to global climate and biodiversity targets. Policymakers should embed urban forests within National Adaptation Plans (NAPs), Nationally Determined Contributions (NDCs), National Biodiversity Strategies and Action Plans (NBSAPs), and Nature Restoration Plans (NRPs) to achieve policy coherence and reduce reporting and implementation costs.

Securing dedicated, long-term finance for stewardship, maintenance, and monitoring is essential. Globally, nature-based solutions address critical challenges such as climate mitigation and strengthening community resilience to environmental hazards [9]. However, funding often prioritises tree planting, while the long-term costs of maintenance, monitoring,

and survival are frequently overlooked. Urban forests, as nature-based solutions, remain severely underfunded, with estimates indicating a need for over $500 billion annually to meet climate and biodiversity goals [107]. Innovative financing through green bonds, biodiversity credits, and carbon markets can supplement public budgets. Incorporating urban forests into ecosystem accounting provides evidence to support investment decisions. Initiatives like Tree Cities of the World, Trees in Cities Challenge, and the Sustainable Forestry Initiative's community standards provide valuable governance models, promoting municipal action and global coordination [108–110]. While public funding dominates, private sector investment mechanisms are key to complementing municipal efforts.

Multi-level governance and active engagement of stakeholders, from community groups to scientists and urban planners, are vital to implement these global commitments (Fig 3). Local governments, community organisations, and residents can jointly support and engage in urban forest stewardship. Adoption of blue-green infrastructure performance standards and municipal level ecosystem accounting will embed forests as central to resilient, equitable urban development [111,112]. As urban forests gain recognition as multifunctional infrastructure, rigorous legal protections and transparent monitoring systems must sustain progress toward equity, climate, and biodiversity objectives.

Key targeted recommendations for mainstreaming urban forests in climate and biodiversity governance are summarised in **Box 3**.

4. **Strengthen resilience through evidence-based management and continuous learning**

Governments must invest in integrated urban forest management systems grounded in comprehensive planning, cross-sectoral collaboration, transparent implementation, and long-term stewardship [113]. Community engagement, including incorporation of Indigenous and rural traditional knowledge, enhances relevance and equity [96]. At the local level, ambitious and measurable targets should extend beyond canopy cover to include 3D forest structure metrics (e.g., tree volume, biomass), habitat connectivity, biodiversity conservation, and equitable access to quality greenspaces [77,86,114]. Transparent tracking via robust, cost-effective tools, such as remote sensing, validated by independent audits, can enable evidence-based management and accountability [115]. Furthermore, artificial intelligence (AI) is increasingly vital for managing urban forest data, integrating information from remote sensing, sensors, and community science to improve tree monitoring and ecosystem assessments [116]. AI-driven tools enable more accurate and cost-effective urban forest management, especially benefiting cities with limited resources, and help amplify social and health benefits by supporting evidence-based stewardship [117,118]. Importantly, learning from past urban forest initiatives is crucial to avoid repeated failures. However, many cities lack systematic records of tree planting success, limiting assessment and evidence-based management [119]. Improved data management and monitoring of planting outcomes will boost long-term urban forest resilience.

Establishing baselines through comprehensive monitoring of urban forest condition, including size-class distribution, species and functional diversity, and structural and ecological attributes, is essential for understanding trends and informing management. However, establishing and maintaining robust local databases to support such monitoring and research remains a major challenge. Although these systems are critical for decision-making [120], many cities, particularly in the rapidly developing Global South, still lack the technical capacity and financial resources to develop and sustain them [121].

Building long-term capacity through international collaboration, knowledge exchange, and investment in local expertise is central to equitable global progress [122]. National and non-profit organisations can create centralised hubs for standardised urban forest data collection and sharing. While many cities maintain local inventories, updating and consistent monitoring remain under-resourced [123–125]. Monitoring must extend to trees on private lands, which often represent substantial urban canopy yet are under-documented despite their contributions to species diversity and structure [73,125,126].

Policymakers should also recognise trade-offs associated with urban forests, including irrigation demands conflicting with sustainable water management, spatial competition with mobility infrastructure, and potential health risks for



Fig 3. Examples of multi-level governance and stewardship initiatives around urban forests in Greater Sydney, Australia. (A) A flexible 'tree hugger' with positive messaging of services provided by street trees during 'Love your trees' week in the City of Willoughby. (B) A sign with a QR code to learn more about new street tree plantings in the City of Sydney. (C) A sign on newly planted trees in the City of Parramatta, notifying the public that the trees are part of an offset program compensating for tree removal for a new light rail line. (D) A free plant giveaway, including trees, for ratepayers of Hornsby Shire. (E) Community training in best-practice tree planting in Blacktown. Images provided by SP.

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vulnerable populations [46,63,127]. Developing prioritisation tools that integrate spatial data on ecosystem services, disservices, equity, tree diversity, urban heat islands, costs, and community needs will support tailored, transparent, and evidence-driven decision-making [128,129].

Diversification of tree species supports climate adaptation and resilience as well as other ecosystem services [130]. Overreliance on narrow species palettes increases vulnerability to climate extremes, pests and pathogens, as evidenced by historic monoculture losses [131,132]. Urban forest policies should adopt functional and genetic diversity, combining species with complementary traits, such as drought and pest resistance, to enhance the resilience of urban areas [130]. This diversity extends to understory vegetation and soil biota, forming multi-layered, functional urban forests. Importantly, native species richness provides a buffer against ecological risks, though climate change and social factors often require the integration of carefully selected and rigorously risk-assessed non-native species for functional complementarity, based on local and site considerations [133–135]. Importantly, species selection and urban forest diversification must be grounded in site assessments and local preferences [22], as environmental conditions, social contexts, and urban gradients vary widely not only between regions but also within cities. This spatial variability influences tree performance, resilience, and ecosystem service provision, needing localised evaluation and adaptive management to ensure successful urban forest outcomes. Balancing diversification with respect for cultural values and Indigenous and local knowledge is essential [97]. Investment in nursery and supply chains that produce a broad range of high-quality, climate-resilient species supports this goal [63].

Finally, guided natural regeneration, a cost-effective restoration method supported by natural seed banks, can accelerate canopy development, reduce planting demands, and increase species diversity [136–139]. Effective management of competing vegetation and protection from disturbance can enable rapid regeneration with minimal inputs [140,141].

Key targeted recommendations for strengthening resilience through evidence-based management and continuous learning are summarised in **Box 4**.

Conclusions

Urban forests are among the most effective nature-based solutions available to cities. Despite their importance, many global policies aimed at protecting urban forests remain underfunded (especially for maintenance and monitoring), and inadequately enforced, limiting much-needed climate action. Thus, immediate investment and safeguarding of urban forests will secure long-term benefits. We have described four urgent policy priorities for immediate action to safeguard and enhance urban forests: investing in urban forests as essential living infrastructure; delivering equitable access to greenspace; mainstreaming urban forests into climate and biodiversity governance frameworks; and strengthening resilience through evidence-based management and continuous learning. These priorities form the foundation for effective urban forest stewardship that can support significant climate, environmental, health, and socio-economic benefits. As urbanisation and climate pressures intensify, urban forests offer a vital pathway to sustain greenspaces and ecosystem functions at a global scale. Greening cities is not just a climate mitigation strategy but a necessity for resilience, health, and social equity for current and future generations. Recognising and investing in urban forests as essential and critical living infrastructure is urgent to secure a sustainable future for Earth's billions of urban residents.

Box 1. Targeted recommendations to enhance urban forest conservation, stewardship, and planning, aiming to protect mature trees, expand canopy cover, and integrate living infrastructure into broader municipal strategies.

1. Establish clear legal frameworks protecting trees on both public and private land, including private tree bylaws that restrict unauthorised removals and mandate minimum levels of urban nature provision on private parcels.

2. Strengthen enforcement and deterrence through advanced monitoring tools such as remote sensing to detect canopy loss at fine spatial scales, paired with tiered penalties for repeat or large-scale illegal tree removal, including possible criminal sanctions where appropriate.
3. Reinvest fines and reward stewardship by directing penalties collected from illegal removals back into urban forestry programs and providing incentives for property owners to conserve mature trees, such as property tax reductions linked to canopy cover.
4. Integrate urban forest conservation into strategic urban planning to prevent conflicts with infrastructure development, preserve mature trees and soils, and limit land-use incentives that favour deforestation over stewardship.
5. Differentiate restoration planting from maintenance planting, with restoration focused on degraded sites and maintenance planting aimed at sustaining mature tree populations, each supported by dedicated budgets.
6. Capitalise on large infrastructure projects as cost-effective opportunities to expand urban tree canopy and diversity through coordinated tree planting.
7. Facilitate comprehensive tree risk assessments and expand knowledge of mitigation techniques to safely extend the lifespan of valuable urban trees.
8. Explicitly include urban forests in climate risk, pandemic preparedness, and critical infrastructure planning, incorporating tailored investment, depreciation, and risk management strategies to address environmental stresses.
9. Set ambitious, measurable, and transparent urban forestry targets that include 3D forest structure, habitat connectivity, biodiversity conservation, and equitable access to greenspace.
10. Incorporate soil health and habitat quality monitoring as integral components of urban forestry initiatives to enhance long-term ecosystem resilience and align urban forests with essential living infrastructure.

Box 2. Targeted recommendations to promote equity and access in urban forestry.

1. Develop site-specific canopy cover, diversity, and greenspace targets co-designed with local communities, ensuring representation of diverse cultural values and needs.
2. Design and implement multilingual public outreach and engagement programs that empower marginalised and underrepresented populations to participate in urban forest stewardship.
3. Ensure meaningful inclusion of marginalised communities in urban forest planning, and decision-making processes through accessible forums and resource provision.
4. Coordinate urban greening efforts with housing, social, and urban planning departments to mitigate green gentrification and promote affordable, inclusive access to ecosystem services.
5. Prioritise equitable tree planting and maintenance in socio-economically disadvantaged neighbourhoods, with transparent monitoring and reporting of progress.
6. Expand and diversify the urban forestry workforce to reflect community demographics, improving cultural competency and representativeness in stewardship roles.
7. Promote partnerships among governments, community organisations, and Indigenous groups to support co-management and culturally inclusive urban forest governance.

8. Promote equitable distribution of tree benefits by explicitly tracking canopy cover, species composition, and ecosystem service supply by neighbourhood socioeconomic status and ethnicity, and using these data to guide investment.
9. Support community or Indigenous stewardship initiatives with dedicated funding, technical resources, and long-term institutional backing, recognising them as core partners in urban forest governance and management.
10. Explicitly monitor and report on green gentrification risks when implementing new greening projects, linking canopy expansion with housing-stability and affordability measures.

Box 3. Targeted recommendations to mainstream urban forests in climate and biodiversity governance.

1. Establish city-level urban forest governance platforms that coordinate climate, biodiversity, and urban planning agencies, as well as community stakeholders, to align urban forest strategies with national and international commitments.
2. Mainstream urban forests into national climate and biodiversity reporting, ensuring that urban canopy, ecosystem services, and equity metrics are systematically included in National Adaptation Plans (NAPs), Nationally Determined Contributions (NDCs), National Biodiversity Strategies and Action Plans (NBSAPs), Nature Restoration Plans (NRPs), and national inventories to track progress and accountability.
3. Develop and enforce minimum urban forest performance standards for municipalities, setting thresholds for canopy cover, species diversity, and equitable access, and requiring regular public reporting to maintain transparency and policy coherence.
4. Embed urban forests into Regional and National Adaptation Plans, Nationally Determined Contributions, and Biodiversity Strategies to align local actions with global climate and biodiversity goals.
5. Secure dedicated long-term financing through public budgets, green bonds, biodiversity credits, and carbon markets.
6. Align financial policies to eliminate incentives for urban forest degradation and promote investments recognising urban forests as essential living infrastructure.
7. Promote multi-level governance approaches engaging local communities, scientists, planners, and policymakers.
8. Integrate urban forests into municipal ecosystem accounting and blue-green infrastructure standards.
9. Use global initiatives (e.g., Tree Cities of the World, Trees in Cities Challenge) as frameworks for coordination and knowledge sharing.
10. Promote communications strategies that highlight urban forest benefits to catalyse private-sector investment.

Box 4. Targeted recommendations to enhance resilience through evidence-based urban forest management.

1. Invest in integrated urban forest management systems that combine cross-sector collaboration, transparent implementation, long-term stewardship, and active community and Indigenous Knowledge engagement.

2. Develop and maintain standardised urban forest monitoring systems and databases at local and regional scales, supported by national or international hubs, incorporating canopy cover, species diversity, tree vitality, and structural attributes through field data, remote sensing, and independent field-based audits.
3. Expand capacity building in rapidly growing and under-resourced cities, especially in the Global South, through international partnerships and knowledge exchange programs.
4. Include private lands in urban forest inventories and monitoring to capture the full extent and diversity of urban canopy cover.
5. Establish ambitious, measurable, and transparent urban forestry targets that incorporate 3D forest structure, habitat connectivity, biodiversity conservation, and equitable access to greenspace.
6. Develop prioritisation and decision-support tools that integrate spatial data on ecosystem services and disservices, equity, costs, and community needs to guide transparent, evidence-based management.
7. Promote species and genetic diversification, incorporating native and carefully selected non-native species with complementary functional traits to build ecosystem resilience.
8. Invest in nursery and supply-chain capacity to produce diverse, high-quality, climate-ready tree species aligned with local ecological and cultural values.
9. Implement and scale guided spontaneous regeneration and support adaptive management through continuous monitoring of planting success, survival, and failure, with feedback loops to improve long-term resilience.
10. Communicate the complexities and trade-offs of urban forest management transparently to support informed decision-making among policymakers and stakeholders.

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References

1. United Nations. United Nations: The world's cities in 2018. Department of Economic and Social Affairs, Population Division. 2018.
2. Lenton TM, Xu C, Abrams JF, Ghadiali A, Loriani S, Sakschewski B, et al. Quantifying the human cost of global warming. Nat Sustain. 2023;6(10):1237–47. https://doi.org/10.1038/s41893-023-01132-6

3. Escobedo FJ, Giannico V, Jim CY, Sanesi G, Lafortezza R. Urban forests, ecosystem services, green infrastructure and nature-based solutions: Nexus or evolving metaphors? Urban Forestry Urban Greening. 2019;37:3–12. https://doi.org/10.1016/j.ufug.2018.02.011
4. Steele W, Handmer J, McShane I. Hot Cities: A Transdisciplinary Agenda. Glos, UK: Edward Elgar Publishing. 2023.
5. Seto KC, Güneralp B, Hutyra LR. Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools. Proc Natl Acad Sci U S A. 2012;109(40):16083–8. https://doi.org/10.1073/pnas.1211658109 PMID: 22988086
6. Grimmond S. Urbanization and global environmental change: Local effects of urban warming. The Geographical Journal. 2007;173(1):83–8.
7. Esperon-Rodriguez M, Arndt S, Asibey MO, Augustinus BA, Bach A, Ballinas M, et al. Urban forests as essential infrastructure for climate resilience and biodiversity: A call to policymakers. PLANTS, PEOPLE, PLANET. 2025;n/a(n/a). https://doi.org/10.1002/ppp3.70125
8. FAO. Guidelines on urban and peri-urban forestry. Rome: Food and Agriculture Organization of the United Nations. 2016.
9. Cohen-Shacham E, Walters G, Janzen C, Maginnis S. Nature-based solutions to address global societal challenges. IUCN, editor. Switzerland: Gland; 2016. 2036.
10. McPhearson T, Kabisch N, Frantzeskaki N. Nature-Based Solutions for Cities. Edward Elgar Publishing. 2023.
11. Cueva J, Yakouchenkova IA, Fröhlich K, Dermann AF, Dermann F, Köhler M, et al. Synergies and trade-offs in ecosystem services from urban and peri-urban forests and their implication to sustainable city design and planning. Sustainable Cities and Society. 2022;82:103903. https://doi.org/10.1016/j.scs.2022.103903
12. Duinker P, Ordóñez C, Steenberg J, Miller K, Toni S, Nitoslawski S. Trees in Canadian Cities: Indispensable Life Form for Urban Sustainability. Sustainability. 2015;7(6):7379–96. https://doi.org/10.3390/su7067379
13. Gómez-Baggethun E, Barton DN. Classifying and valuing ecosystem services for urban planning. Ecological Economics. 2013;86:235–45. https://doi.org/10.1016/j.ecolecon.2012.08.019
14. Livesley SJ, McPherson GM, Calfapietra C. The Urban Forest and Ecosystem Services: Impacts on Urban Water, Heat, and Pollution Cycles at the Tree, Street, and City Scale. J Environ Qual. 2016;45(1):119–24. https://doi.org/10.2134/jeq2015.11.0567 PMID: 26828167
15. Keeler BL, Hamel P, McPhearson T, Hamann MH, Donahue ML, Meza Prado KA, et al. Social-ecological and technological factors moderate the value of urban nature. Nat Sustain. 2019;2(1):29–38. https://doi.org/10.1038/s41893-018-0202-1
16. UN. System of Environmental Economic Accounting – Ecosystem Accounting. New York: United Nations; 2021.
17. Alvey AA. Promoting and preserving biodiversity in the urban forest. Urban Forestry & Urban Greening. 2006;5(4):195–201. https://doi.org/10.1016/j.ufug.2006.09.003
18. Wickramathilaka N, Ujang U, Azri S, Choon TL. Influence of urban green spaces on road traffic noise levels: - A review. Int Arch Photogramm Remote Sens Spatial Inf Sci. 2022;XLVIII-4/W3-2022:195–201. https://doi.org/10.5194/isprs-archives-xlviii-4-w3-2022-195-2022
19. Sharmin M, Tjoelker MG, Pfautsch S, Esperon-Rodriguez M, Rymer PD, Power SA. Tree crown traits and planting context contribute to reducing urban heat. Urban Forestry & Urban Greening. 2023;83:127913. https://doi.org/10.1016/j.ufug.2023.127913
20. Sharmin M, Tjoelker MG, Pfautsch S, Esperón-Rodriguez M, Rymer PD, Power SA. Tree Traits and Microclimatic Conditions Determine Cooling Benefits of Urban Trees. Atmosphere. 2023;14(3):606. https://doi.org/10.3390/atmos14030606
21. Ordóñez C, Duinker PN. Assessing the vulnerability of urban forests to climate change. Environ Rev. 2014;22(3):311–21. https://doi.org/10.1139/er-2013-0078
22. Brandt L, Derby Lewis A, Fahey R, Scott L, Darling L, Swanston C. A framework for adapting urban forests to climate change. Environmental Science & Policy. 2016;66:393–402. https://doi.org/10.1016/j.envsci.2016.06.005
23. Ordóñez Barona C. Adopting public values and climate change adaptation strategies in urban forest management: A review and analysis of the relevant literature. J Environ Manage. 2015;164:215–21. https://doi.org/10.1016/j.jenvman.2015.09.004 PMID: 26410091
24. Carrus G, Dadvand P, Sanesi G. The role and value of urban forests and green infrastructure in promoting human health and wellbeing. In: Pearlmutter D, Calfapietra C, Samson R, O'Brien L, Krajter Ostoić S, Sanesi G, ed. The urban forest: cultivating green infrastructure for people and the environment. Cham: Springer International Publishing. 2017. 217–30.
25. Van den Bosch M. Impacts of urban forests on physical and mental health and wellbeing. Routledge handbook of urban forestry. Routledge. 2017. 82–95.
26. Endreny TA, Ciolfi M, Endreny A, Chiocchini F, Calfapietra C. Neighborhood-scale reductions in heatwave burden projected under a 30% minimum tree cover scenario. npj Urban Sustain. 2025;5(1). https://doi.org/10.1038/s42949-025-00219-7
27. Russo A, Escobedo FJ, Cirella GT, Zerbe S. Edible green infrastructure: An approach and review of provisioning ecosystem services and disservices in urban environments. Agriculture, Ecosystems & Environment. 2017;242:53–66. https://doi.org/10.1016/j.agee.2017.03.026
28. Kowalski JM, Conway TM. The routes to fruit: Governance of urban food trees in Canada. Urban Forestry & Urban Greening. 2023;86:128045. https://doi.org/10.1016/j.ufug.2023.128045
29. Nesbitt L, Hotte N, Barron S, Cowan J, Sheppard SRJ. The social and economic value of cultural ecosystem services provided by urban forests in North America: A review and suggestions for future research. Urban Forestry & Urban Greening. 2017;25:103–11. https://doi.org/10.1016/j.ufug.2017.05.005

30. Gopal D, von der Lippe M, Kowarik I. Sacred sites as habitats of culturally important plant species in an Indian megacity. Urban Forestry & Urban Greening. 2018;32:113–22. https://doi.org/10.1016/j.ufug.2018.04.003
31. Smith IA, Dearborn VK, Hutyra LR. Live fast, die young: Accelerated growth, mortality, and turnover in street trees. PLoS One. 2019;14(5):e0215846. https://doi.org/10.1371/journal.pone.0215846 PMID: 31067257
32. Haase D, Hellwig R. Effects of heat and drought stress on the health status of six urban street tree species in Leipzig, Germany. Trees, Forests and People. 2022;8:100252. https://doi.org/https://doi.org/10.1016/j.tfp.2022.100252
33. Marchin RM, Esperon-Rodriguez M, Tjoelker MG, Ellsworth DS. Crown dieback and mortality of urban trees linked to heatwaves during extreme drought. Sci Total Environ. 2022;850:157915. https://doi.org/10.1016/j.scitotenv.2022.157915 PMID: 35944640
34. Equiza MA, Calvo-Polanco M, Cirelli D, Señorans J, Wartenbe M, Saunders C. Long-term impact of road salt (NaCl) on soil and urban trees in Edmonton, Canada. Urban Forestry & Urban Greening. 2017;21:16–28. https://doi.org/https://doi.org/10.1016/j.ufug.2016.11.003
35. Pandey B, Ghosh A. Urban ecosystem services and climate change: a dynamic interplay. Front Sustain Cities. 2023;5. https://doi.org/10.3389/frsc.2023.1281430
36. Esperon-Rodriguez M, Tjoelker MG, Lenoir J, Baumgartner JB, Beaumont LJ, Nipperess DA, et al. Climate change increases global risk to urban forests. Nat Clim Chang. 2022;12(10):950–5. https://doi.org/10.1038/s41558-022-01465-8
37. Tubby KV, Webber JF. Pests and diseases threatening urban trees under a changing climate. Forestry. 2010;83(4):451–9. https://doi.org/10.1093/forestry/cpq027
38. Paap T, Burgess TI, Wingfield MJ. Urban trees: bridge-heads for forest pest invasions and sentinels for early detection. Biol Invasions. 2017;19(12):3515–26. https://doi.org/10.1007/s10530-017-1595-x
39. Diana Grecia A-M, Sergio Arturo T-S, Marlenne G-R. Review: Implications of Air Pollution on Trees Located in Urban Areas. Earth. 2025;6(2):38. https://doi.org/10.3390/earth6020038
40. van Vliet J. Direct and indirect loss of natural area from urban expansion. Nat Sustain. 2019;2(8):755–63. https://doi.org/10.1038/s41893-019-0340-0
41. Locosselli GM, Camargo de EP, Moreira TCL, Todesco E, Andrade de MF, André de CDS, et al. The role of air pollution and climate on the growth of urban trees. Science of The Total Environment. 2019;666:652–61. https://doi.org/10.1016/j.scitotenv.2019.02.291
42. Sanesi G, Colangelo G, Lafortezza R, Calvo E, Davies C. Urban green infrastructure and urban forests: a case study of the Metropolitan Area of Milan. Landscape Research. 2016;42(2):164–75. https://doi.org/10.1080/01426397.2016.1173658
43. Raupp M, Cumming A, Raupp E. Street Tree Diversity in Eastern North America and Its Potential for Tree Loss to Exotic Borers. AUF. 2006;32(6):297–304. https://doi.org/10.48044/jauf.2006.038
44. Nowak DJ, Greenfield EJ. The increase of impervious cover and decrease of tree cover within urban areas globally (2012–2017). Urban Forestry & Urban Greening. 2020;49:126638. https://doi.org/10.1016/j.ufug.2020.126638
45. Hilbert D, Roman L, Koeser A, Vogt J, van Doorn N. Urban Tree Mortality: A Literature Review. AUF. 2019;45(5). https://doi.org/10.48044/jauf.2019.015
46. Roman LA, Conway TM, Eisenman TS, Koeser AK, Ordóñez Barona C, Locke DH, et al. Beyond "trees are good": Disservices, management costs, and tradeoffs in urban forestry. Ambio. 2021;50(3):615–30. https://doi.org/10.1007/s13280-020-01396-8 PMID: 33011917
47. Cariñanos P, Casares-Porcel M. Urban green zones and related pollen allergy: A review. Some guidelines for designing spaces with low allergy impact. Landscape and Urban Planning. 2011;101(3):205–14. https://doi.org/10.1016/j.landurbplan.2011.03.006
48. Kasprzyk I, Ćwik A, Kluska K, Wójcik T, Cariñanos P. Allergenic pollen concentrations in the air of urban parks in relation to their vegetation. Urban Forestry & Urban Greening. 2019;46:126486. https://doi.org/10.1016/j.ufug.2019.126486
49. Calfapietra C, Fares S, Manes F, Morani A, Sgrigna G, Loreto F. Role of Biogenic Volatile Organic Compounds (BVOC) emitted by urban trees on ozone concentration in cities: a review. Environ Pollut. 2013;183:71–80. https://doi.org/10.1016/j.envpol.2013.03.012 PMID: 23597803
50. Venter ZS, Hassani A, Stange E, Schneider P, Castell N. Reassessing the role of urban green space in air pollution control. Proc Natl Acad Sci U S A. 2024;121(6):e2306200121. https://doi.org/10.1073/pnas.2306200121 PMID: 38285938
51. Cariñanos P, Calaza-Martínez P, O'Brien L, Calfapietra C. The cost of greening: disservices of urban trees. In: Pearlmutter D, Calfapietra C, Samson R, O'Brien L, Krajter Ostoic S, Sanesi G, ed. The Urban Forest Cultivating Green Infrastructure for People and the Environment. Future City 7: Springer. 2017. 79–87.
52. Masini E, Tomao A, Corona P, Fattorini L, Giuliarelli D, Portoghesi L, et al. The ecosystem disservices of trees on sidewalks: A study based on a municipality urban tree inventory in Central Italy. Urban Forestry & Urban Greening. 2023;86:128007. https://doi.org/10.1016/j.ufug.2023.128007
53. Vogt J, Hauer R, Fischer B. The Costs of Maintaining and Not Maintaining the Urban Forest: A Review of the Urban Forestry and Arboriculture Literature. AUF. 2015;41(6). https://doi.org/10.48044/jauf.2015.027
54. Maes MJA, Jones KE, Toledano MB, Milligan B. Mapping synergies and trade-offs between urban ecosystems and the sustainable development goals. Environmental Science & Policy. 2019;93:181–8. https://doi.org/10.1016/j.envsci.2018.12.010
55. Ghalehteimouri KJ, Ros FC, Rambat S. Flood risk assessment through rapid urbanization LULC change with destruction of urban green infrastructures based on NASA Landsat time series data: A case of study Kuala Lumpur between 1990–2021. Ecological Frontiers. 2024;44(2):289–306. https://doi.org/10.1016/j.chnaes.2023.06.007

56. Dale AG, Frank SD. Warming and drought combine to increase pest insect fitness on urban trees. PLoS One. 2017;12(3):e0173844. https://doi.org/10.1371/journal.pone.0173844 PMID: 28278206
57. Huang S, Wang S, Gan Y, Wang C, Horton DE, Li C, et al. Widespread global exacerbation of extreme drought induced by urbanization. Nat Cities. 2024;1(9):597–609. https://doi.org/10.1038/s44284-024-00102-z
58. Tee SL, Samantha LD, Kamarudin N, Akbar Z, Lechner AM, Ashton-Butt A, et al. Urban forest fragmentation impoverishes native mammalian biodiversity in the tropics. Ecol Evol. 2018;8(24):12506–21. https://doi.org/10.1002/ece3.4632 PMID: 30619561
59. Nowak DJ, Greenfield EJ. Declining urban and community tree cover in the United States. Urban Forestry & Urban Greening. 2018;32:32–55. https://doi.org/10.1016/j.ufug.2018.03.006
60. Fu Q, Zheng Z, Sarker MNI, Lv Y. Combating urban heat: Systematic review of urban resilience and adaptation strategies. Heliyon. 2024;10(17):e37001. https://doi.org/10.1016/j.heliyon.2024.e37001 PMID: 39281560
61. Carrus G, Scopelliti M, Lafortezza R, Colangelo G, Ferrini F, Salbitano F, et al. Go greener, feel better? The positive effects of biodiversity on the well-being of individuals visiting urban and peri-urban green areas. Landscape and Urban Planning. 2015;134:221–8. https://doi.org/10.1016/j.landurbplan.2014.10.022
62. Richards DR, Belcher RN. Global Changes in Urban Vegetation Cover. Remote Sensing. 2019;12(1):23. https://doi.org/10.3390/rs12010023
63. Esperon-Rodriguez M, Gallagher R, Calfapietra C, Cariñanos P, Dobbs C, Eleuterio AA, et al. Barriers and opportunities for resilient and sustainable urban forests. Nat Cities. 2025;2(4):290–8. https://doi.org/10.1038/s44284-025-00212-2
64. Phillips A, Canters F, Khan AZ. Analyzing spatial inequalities in use and experience of urban green spaces. Urban Forestry & Urban Greening. 2022;74:127674. https://doi.org/10.1016/j.ufug.2022.127674
65. Bressane A, Cunha Pinto da JP, Castro Medeiros de LC. Urban green space disparities: Implications of environmental injustice for public health. Urban Forestry & Urban Greening. 2024;99:128441. https://doi.org/10.1016/j.ufug.2024.128441
66. Visintin C, Garrard GE, Weisser WW, Baracco M, Hobbs RJ, Bekessy SA. Designing cities for everyday nature. Conserv Biol. 2025;39(1):e14328. https://doi.org/10.1111/cobi.14328 PMID: 39045810
67. Martin AJF, Fleming A, Conway TM. Distributional inequities in tree density, size, and species diversity in 32 Canadian cities. npj Urban Sustain. 2025;5(1). https://doi.org/10.1038/s42949-025-00210-2
68. Diep L, McPhearson T. Empowering cities globally: Four levers for transformative urban adaptation with nature-based solutions. Proc Natl Acad Sci U S A. 2025;122(29):e2315912121. https://doi.org/10.1073/pnas.2315912121 PMID: 40658839
69. Grabowski ZJ, McPhearson T, Pickett STA. Transforming US urban green infrastructure planning to address equity. Landscape and Urban Planning. 2023;229:104591. https://doi.org/10.1016/j.landurbplan.2022.104591
70. Sharmin M, Power SA, Nguyen M, Rymer PD, Tjoelker MG, Esperon-Rodriguez M. Climate and socio-economic drivers of urban tree abundance, richness and composition in Australian cities. Cities. 2026;169:106532. https://doi.org/10.1016/j.cities.2025.106532
71. Ziter CD, Buxton RT. Anti-racist, decolonial, and transdisciplinary approaches for nature-based solutions that benefit biodiversity and human well-being. Environ Res: Ecology. 2025;4(2):020201. https://doi.org/10.1088/2752-664x/addd27
72. Nesticò A, Guarini MR, Morano P, Sica F. An Economic Analysis Algorithm for Urban Forestry Projects. Sustainability. 2019;11(2):314. https://doi.org/10.3390/su11020314
73. Sousa-Silva R, Lambry T, Cameron E, Belluau M, Paquette A. Urban forests – Different ownership translates to greater diversity of trees. Urban Forestry & Urban Greening. 2023;88:128084. https://doi.org/10.1016/j.ufug.2023.128084
74. Bassett CG. Bridging gaps in urban forest management to achieve urban sustainability goals. 2024.
75. Wirtz Z, Hagerman S, Hauer RJ, Konijnendijk CC. What makes urban forest governance successful? – A study among Canadian experts. Urban Forestry & Urban Greening. 2021;58:126901. https://doi.org/10.1016/j.ufug.2020.126901
76. Russo A, Escobedo FJ. From Smart Urban Forests to Edible Cities: New Approaches in Urban Planning and Design. UP. 2022;7(2):131–4. https://doi.org/10.17645/up.v7i2.5804
77. Morgenroth J, Doick K, Hauer R, Locke DH, Barona CO, Roman LA, et al. Urban tree cover targets: The good, the bad and the SMART. Urban Forestry & Urban Greening. 2025;112:128979. https://doi.org/10.1016/j.ufug.2025.128979
78. Young RF. Planting the Living City. Journal of the American Planning Association. 2011;77(4):368–81. https://doi.org/10.1080/01944363.2011.616996
79. Mell IC, Henneberry J, Hehl-Lange S, Keskin B. Promoting urban greening: Valuing the development of green infrastructure investments in the urban core of Manchester, UK. Urban Forestry & Urban Greening. 2013;12(3):296–306. https://doi.org/10.1016/j.ufug.2013.04.006
80. De la Sota C, Ruffato-Ferreira VJ, Ruiz-García L, Alvarez S. Urban green infrastructure as a strategy of climate change mitigation. A case study in northern Spain. Urban Forestry & Urban Greening. 2019;40:145–51. https://doi.org/10.1016/j.ufug.2018.09.004
81. Macci C, Vannucchi F, Scartazza A, Masciandaro G, Doni S, Peruzzi E. Soil–Plant Indicators for Assessing Nutrient Cycling and Ecosystem Functionality in Urban Forestry. Urban Science. 2025;9(3):82. https://doi.org/10.3390/urbansci9030082
82. Steenberg JWN, Millward AA, Nowak DJ, Robinson PJ, Ellis A. Forecasting Urban Forest Ecosystem Structure, Function, and Vulnerability. Environmental Management. 2017;59(3):373–92. https://doi.org/10.1007/s00267-016-0782-3

83. Lüttge U, Buckeridge M. Trees: structure and function and the challenges of urbanization. Trees. 2020;37(1):9–16. https://doi.org/10.1007/s00468-020-01964-1
84. Sharmin M, Tjoelker MG, Esperon-Rodriguez M, Katlav A, Gilpin A-M, Rymer PD, et al. Urban greening with shrubs can supercharge invertebrate abundance and diversity. Sci Rep. 2024;14(1):8735. https://doi.org/10.1038/s41598-024-58909-8 PMID: 38627432
85. Day S, Wiseman PE, Dickinson S, Harris JR. Tree Root Ecology in the Urban Environment and Implications for a Sustainable Rhizosphere. AUF. 2010;36(5):193–205. https://doi.org/10.48044/jauf.2010.026
86. Russo A, Cirella GT. Urban Ecosystem Services in a Rapidly Urbanizing World: Scaling up Nature's Benefits from Single Trees to Thriving Urban Forests. Land. 2024;13(6):786. https://doi.org/10.3390/land13060786
87. Ordóñez Barona C, Jain A, Heppner M, St Denis A, Boyer D, Lane J, et al. Gaps in the implementation of urban forest management plans across canadian cities. Landscape and Urban Planning. 2024;251:105168. https://doi.org/10.1016/j.landurbplan.2024.105168
88. Thompson CW, Roe J, Aspinall P, Zuin A, Travlou P, Bell S. Community green: using local spaces to tackle inequality and improve health. London. 2010.
89. Wüstemann H, Kalisch D, Kolbe J. Access to urban green space and environmental inequalities in Germany. Landscape and Urban Planning. 2017;164:124–31. https://doi.org/10.1016/j.landurbplan.2017.04.002
90. Ordóñez-Barona C. How different ethno-cultural groups value urban forests and its implications for managing urban nature in a multicultural landscape: A systematic review of the literature. Urban Forestry & Urban Greening. 2017;26:65–77. https://doi.org/10.1016/j.ufug.2017.06.006
91. Esperon-Rodriguez M, Bond J, Esperon Rodriguez D, Jeffries P, Van Ermel Scherer S, Tjoelker MG. Aboriginal and Torres Strait islander people's perceptions and preferences of urban forests in Australia. Discov Cities. 2025;2(1). https://doi.org/10.1007/s44327-025-00079-x
92. Gerstenberg T, Hofmann M. Perception and preference of trees: A psychological contribution to tree species selection in urban areas. Urban Forestry & Urban Greening. 2016;15:103–11. https://doi.org/10.1016/j.ufug.2015.12.004
93. Russo A, Esperon-Rodriguez M, St-Denis A, Tjoelker MG. Native vs. Non-Native Plants: Public Preferences, Ecosystem Services, and Conservation Strategies for Climate-Resilient Urban Green Spaces. Land. 2025;14(5):954. https://doi.org/10.3390/land14050954
94. Scott JL, Tenneti A. Race in nature stewardship: an autoethnography of two racialised volunteers in urban ecology. Environ Res: Ecology. 2024;3(3):035006. https://doi.org/10.1088/2752-664x/ad7033
95. Mata L, Ramalho CE, Kennedy J, Parris KM, Valentine L, Miller M, et al. Bringing nature back into cities. People and Nature. 2020;2(2):350–68. https://doi.org/10.1002/pan3.10088
96. Grabowski ZJ, Wijsman K, Tomateo C, McPhearson T. How deep does justice go? Addressing ecological, indigenous, and infrastructural justice through nature-based solutions in New York City. Environmental Science & Policy. 2022;138:171–81. https://doi.org/10.1016/j.envsci.2022.09.022
97. Martin AJF, Bacon ES, Migizikwe N, Soucy S, Grant A, Conway TM. History, engagement, and visibility of Indigenous Peoples in urban forest management plans from Canada and the United States. Environmental Science & Policy. 2025;166:104026. https://doi.org/10.1016/j.envsci.2025.104026
98. Wolf KL. Trees and Business District Preferences: A Case Study of Athens, Georgia, U.S. isa. 2004;30(6):336–46. https://doi.org/10.48044/jauf.2004.041
99. Landry F, Dupras J, Messier C. Convergence of urban forest and socio-economic indicators of resilience: A study of environmental inequality in four major cities in eastern Canada. Landscape and Urban Planning. 2020;202:103856. https://doi.org/10.1016/j.landurbplan.2020.103856
100. McDonald RI, Biswas T, Chakraborty TC, Kroeger T, Cook-Patton SC, Fargione JE. Current inequality and future potential of US urban tree cover for reducing heat-related health impacts. npj Urban Sustain. 2024;4(1). https://doi.org/10.1038/s42949-024-00150-3
101. McDonald RI, Biswas T, Sachar C, Housman I, Boucher TM, Balk D, et al. The tree cover and temperature disparity in US urbanized areas: Quantifying the association with income across 5,723 communities. PLoS One. 2021;16(4):e0249715. https://doi.org/10.1371/journal.pone.0249715 PMID: 33909628
102. Li L, Kross A, Ziter CD, Eicker U. Analyzing spatial patterns of urban green infrastructure for urban cooling and social equity. Urban Forestry & Urban Greening. 2025;112:128983. https://doi.org/10.1016/j.ufug.2025.128983
103. Qi H, Dempsey N, Cameron R. Seeing the forest for the trees? An exploration of the Miyawaki forest method in the UK. Arboricultural Journal. 2024;46(4):292–304. https://doi.org/10.1080/03071375.2024.2394355
104. Agreement P. Report of the conference of the parties to the United Nations framework convention on climate change. HeinOnline. 2015.
105. CBD. The Convention on Biological Diversity. 2025. https://www.cbd.int
106. UN. The 17 Goals. 2025. https://sdgs.un.org/goals
107. State of Finance for Nature in Cities: Summary for Local Policymakers. Paris: UN. 2023.
108. TC. Tree Cities of the World. 2025. https://www.treecitiesoftheworld.org
109. UNECE. Trees in Cities Challenge. 2025. https://land.unece.org/treesincities
110. SFI. Urban and Community Forest Sustainability Standard. 2025. https://forests.org/sfi-urban-forestry-standard2025

111. Stange EE, Barton DN, Andersson E, Haase D. Comparing the implicit valuation of ecosystem services from nature-based solutions in performance-based green area indicators across three European cities. Landscape and Urban Planning. 2022;219:104310. https://doi.org/10.1016/j.landurbplan.2021.104310

112. Babi Almenar J, Marando F, Vallecillo S, Zulian G, Cortinovis C, Zurbaran Nucci M. Urban ecosystem accounts following the SEEA EA standard: A pilot application in Europe. Luxembourg. 2023.

113. Hutt-Taylor K, Chamberland-Fontaine S, Bardekjian AC, Khan A, Duncan A, Mokdad A, et al. Improving cross-sectoral collaboration towards urban nature-based solutions: insights from a participatory workshop. FACETS. 2025;10:1–17. https://doi.org/10.1139/facets-2025-0035

114. Croeser T, Weisser WW, Hurley J, Rötzer T, Parhizgar L, Sun Q (Chayn), et al. Defining 'adequate' tree protection: Meeting urban canopy targets requires careful retention of mature trees. Landscape and Urban Planning. 2025;264:105484. https://doi.org/10.1016/j.landurbplan.2025.105484

115. Venter ZS, Czúcz B, Stange E, Nowell MS, Simensen T, Immerzeel B, et al. 'Uncertainty audit' for ecosystem accounting: Satellite-based ecosystem extent is biased without design-based area estimation and accuracy assessment. Ecosystem Services. 2024;66:101599. https://doi.org/10.1016/j.ecoser.2024.101599

116. Zeybek M. Integrating advanced remote sensing technologies and machine learning in urban forestry: a comprehensive review and future outlook. Meas Sci Technol. 2025;36(6):062004. https://doi.org/10.1088/1361-6501/addf66

117. Salbitano F, Borelli S, Conigliaro M, Yahya NA, Sanesi G, Chen Y. Urban forest benefits in developing and industrializing countries. In: Ferrini F, Konijnendijk CC, Fini A, ed. Routledge handbook of urban forestry. London: Routledge. 2017: 136–51.

118. Pauleit S, Vasquéz A, Maruthaveeran S, Liu L, Cilliers SS. Urban Green Infrastructure in the Global South. In: Shackleton CM, Cilliers SS, Davoren E, du Toit MJ, ed. Urban Ecology in the global South. Cham: Springer International Publishing. 2021: 107–43. https://doi.org/10.1007/978-3-030-67650-6_5

119. Esperon-Rodriguez M, Rymer P, Power S, Barton D, Cariñanos P, Dobbs C. Assessing climate risk to support urban forests in a changing climate. Plants, People, Planet. 2022;4(3):201–13.

120. Booth TH. The Need for a Global Tree Trial Database. New Forests. 2022;54(1):1–7. https://doi.org/10.1007/s11056-022-09940-y

121. Barona CO, Devisscher T, Dobbs C, Aguilar LO, Baptista MD, Navarro NM, et al. Trends in Urban Forestry Research in Latin America & The Caribbean: A Systematic Literature Review and Synthesis. Urban Forestry & Urban Greening. 2020;47:126544. https://doi.org/10.1016/j.ufug.2019.126544

122. Frantzeskaki N, Wijsman K, Kabisch N, McPhearson T. Inter- and transdisciplinary knowledge is critical for nature-based solutions to contribute to just urban transformations. Proc Natl Acad Sci U S A. 2025;122(29):e2315911121. https://doi.org/10.1073/pnas.2315911121 PMID: 40658852

123. Esperon-Rodriguez M, Quintans D, Rymer PD. Urban tree inventories as a tool to assess tree growth and failure: The case for Australian cities. Landscape and Urban Planning. 2023;233:104705. https://doi.org/10.1016/j.landurbplan.2023.104705

124. Ma B, Hauer RJ, Östberg J, Koeser AK, Wei H, Xu C. A global basis of urban tree inventories: What comes first the inventory or the program. Urban Forestry & Urban Greening. 2021;60:127087. https://doi.org/10.1016/j.ufug.2021.127087

125. Östberg J, Wiström B, Randrup TB. The state and use of municipal tree inventories in Swedish municipalities – results from a national survey. Urban Ecosyst. 2018;21(3):467–77. https://doi.org/10.1007/s11252-018-0732-3

126. Hutt-Taylor K, Ziter CD. Private trees contribute uniquely to urban forest diversity, structure and service-based traits. Urban Forestry & Urban Greening. 2022;78:127760. https://doi.org/10.1016/j.ufug.2022.127760

127. Back P, Collins AM. Negotiating the green obstacle course: Ranking priorities and problems for municipal green infrastructure implementation. Urban Forestry & Urban Greening. 2022;67:127436. https://doi.org/10.1016/j.ufug.2021.127436

128. St-Denis A, Maure F, Belbahar R, Delagrange S, Handa IT, Kneeshaw D, et al. An Urban Forest Diversification Software to Improve Resilience to Global Change. Urban Forestry. 2023;50(1):76–91. https://doi.org/10.48044/jauf.2023.027

129. Werbin ZR, Heidari L, Buckley S, Brochu P, Butler LJ, Connolly C, et al. A tree-planting decision support tool for urban heat mitigation. PLoS One. 2020;15(10):e0224959. https://doi.org/10.1371/journal.pone.0224959 PMID: 33031384

130. Paquette A, Sousa-Silva R, Maure F, Cameron E, Belluau M, Messier C. Praise for diversity: A functional approach to reduce risks in urban forests. Urban Forestry & Urban Greening. 2021;62:127157. https://doi.org/10.1016/j.ufug.2021.127157

131. Donovan GH, Butry DT, Michael YL, Prestemon JP, Liebhold AM, Gatziolis D, et al. The relationship between trees and human health: evidence from the spread of the emerald ash borer. Am J Prev Med. 2013;44(2):139–45. https://doi.org/10.1016/j.amepre.2012.09.066 PMID: 23332329

132. Roman LA, Pearsall H, Eisenman TS, Conway TM, Fahey RT, Landry S, et al. Human and biophysical legacies shape contemporary urban forests: A literature synthesis. Urban Forestry & Urban Greening. 2018;31:157–68. https://doi.org/10.1016/j.ufug.2018.03.004

133. Delavaux CS, Crowther TW, Zohner CM, Robmann NM, Lauber T, van den Hoogen J, et al. Native diversity buffers against severity of non-native tree invasions. Nature. 2023;621(7980):773–81. https://doi.org/10.1038/s41586-023-06440-7 PMID: 37612513

134. Schlaepfer MA, Bacher S, Kueffer C. Non-native species and novel ecosystems: ecology, conservation, and management perspectives. Trends in Ecology & Evolution. 2020;35:470–82.

135. Busk KØ, Svenning J-C. A global-change winner? Global expansion potential, ecological drivers and hybridization risk of the disturbance-promoted honey locust (Gleditsia). Discov Plants. 2025;2(1). https://doi.org/10.1007/s44372-025-00358-2

136. Kimball S, Lulow M, Sorenson Q, Balazs K, Fang Y, Davis SJ, et al. Cost-effective ecological restoration. Restoration Ecology. 2015;23(6):800–10. https://doi.org/10.1111/rec.12261

137. Crouzeilles R, Beyer HL, Monteiro LM, Feltran-Barbieri R, Pessôa ACM, Barros FSM, et al. Achieving cost-effective landscape-scale forest restoration through targeted natural regeneration. Conservation letters. 2020;13(3). https://doi.org/10.1111/conl.12709

138. Pettorelli N, Dancer AD, Durant SM, Hoffmann M, Laughlin B, Pilkington J. Rewilding our cities. London, UK. 2022.

139. Esperon-Rodriguez M, Svenning J-C. Urban rewilding as a nature-based solution: a critical perspective. Critical Insights in Plant Science. 2025;1(1). https://doi.org/10.1080/29932297.2026.2629049

140. Dietze MC, Clark JS. Changing the gap dynamics paradigm: vegetative regeneration control on forest response to disturbance. Ecological Monographs. 2008;78(3):331–47. https://doi.org/10.1890/07-0271.1

141. Kozlowski TT. Physiological ecology of natural regeneration of harvested and disturbed forest stands: implications for forest management. Forest Ecology and Management. 2002;158(1–3):195–221. https://doi.org/10.1016/s0378-1127(00)00712-x